A sharp entropic Rogers–Shephard inequality via simplex fibers

Theorem 1. *Let $n \geq 1$, and let X, X' be independent random vectors with the same log-concave probability density f on $\mathbb{R}^n$. Both entropies below are finite, and*

$$\mathrm{h}(X - X') - \mathrm{h}(X) \leq n \log 2. \tag{1}$$

The constant is best possible in every dimension. Equality is attained when X is an invertible affine image of a vector of independent one-sided exponential random variables.

1 Notation

All logarithms are natural, all volumes are Lebesgue volumes in the indicated dimension, and $0 \log 0 = 0$. For a probability density f on $\mathbb{R}^n$, write

$$\mathrm{h}(f) = -\int_{\mathbb{R}^n} f(x) \log f(x)\, dx, \qquad \widetilde{f}(x) = f(-x).$$

We use the usual almost-everywhere convention for densities. A log-concave density may be represented as $f = e^{-V}/Z$, where $V : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is proper, lower semicontinuous and convex, and $Z = \int e^{-V} \in (0, \infty)$. Indeed, a convex log-potential on the interior of its domain can be replaced by its lower semicontinuous closure; the density changes only on the boundary of a full-dimensional convex set, which is a Lebesgue null set.

Set

$$H_k = \sum_{j=1}^{k} \frac{1}{j}, \qquad F(k) = k(H_{2k} - H_k) \quad (k \geq 1). \tag{2}$$

2 An entropy estimate on convex bodies

Lemma 2. *Let $K \subset \mathbb{R}^d$ be a convex body and let $u : K \to [0, 1]$ be a measurable concave function with maximum 1 attained at $x_0 \in K$. Extend u by zero outside K. For $p > 0$, put $I(p) = \int_K u^p$ and $\varphi_p = u^p / I(p)$. Then $I(p) > 0$, $\mathrm{h}(\varphi_p)$ is finite, and*

$$\mathrm{h}(\varphi_p) - \log I(p) \leq p \sum_{j=1}^{d} \frac{1}{p + j}. \tag{3}$$

Proof. For $0 < t < 1$, let $v(t) = |\{x \in K : u(x) > t\}|$. If $0 < s < t < 1$, concavity gives

$$x_0 + \frac{1 - t}{1 - s}(\{u > s\} - x_0) \subset \{u > t\}.$$

Consequently

$$w(t) = \frac{v(t)}{(1 - t)^d} \quad \text{is nondecreasing on } (0, 1). \tag{4}$$

The same homothety applied to K shows $v(t) > 0$ for $t < 1$; in particular $I(p) > 0$. Layer integration gives

$$I(p) = p \int_0^1 t^{p-1} v(t)\, dt = p \int_0^1 t^{p-1} (1 - t)^d w(t)\, dt. \tag{5}$$

Differentiation is legitimate for every $p > 0$: on a compact interval of positive exponents, $u^p|\log u|$ is uniformly bounded, while $|K| < \infty$. The same conclusion follows on the right of (5) from $v(t) \le |K|$ and the integrability of $t^{p-1}|\log t|$.

Let μ_p be the probability measure on $(0,1)$ proportional to $t^{p-1}(1-t)^d\,dt$. Two nondecreasing functions have nonnegative covariance, so (4) yields

$$\frac{\int w(t)\log t\,d\mu_p(t)}{\int w(t)\,d\mu_p(t)} \ge \int \log t\,d\mu_p(t). \tag{6}$$

For completeness, multiply by $2\int w\,d\mu_p$ and expand

$$\iint (w(t)-w(s))(\log t - \log s)\,d\mu_p(t)\,d\mu_p(s) \ge 0.$$

Every term is absolutely integrable by $v \le |K|$ and (5); thus this argument also covers unbounded w.

Repeated integration by parts gives, for $p > 0$ and integer $d \ge 0$,

$$B(p, d+1) := \int_0^1 t^{p-1}(1-t)^d\,dt = \frac{d!}{p(p+1)\cdots(p+d)}. \tag{7}$$

Its logarithmic derivative is

$$\int \log t\,d\mu_p(t) = -\sum_{j=0}^{d} \frac{1}{p+j}.$$

Differentiating (5) and using (6), we obtain

$$\frac{I'(p)}{I(p)} = \frac{1}{p} + \frac{\int w\log t\,d\mu_p}{\int w\,d\mu_p} \ge -\sum_{j=1}^{d}\frac{1}{p+j}.$$

Finally, $\mathrm{h}(\varphi_p) = \log I(p) - pI'(p)/I(p)$, proving (3). Finiteness follows from boundedness and compact support of φ_p. □

Proposition 3. *If Z, Z' are independent and uniform on a convex body $B \subset \mathbb{R}^N$, then*

$$\mathrm{h}(Z - Z') - \mathrm{h}(Z) \le F(N). \tag{8}$$

Proof. Write $v_B = |B| > 0$. The density of $Z - Z'$ is

$$q(z) = \frac{|B \cap (z+B)|}{v_B^2}.$$

For $C_z = B \cap (z+B)$ and $z, w \in B - B$,

$$(1-\lambda)C_z + \lambda C_w \subset C_{(1-\lambda)z+\lambda w} \quad (0 \le \lambda \le 1).$$

Brunn–Minkowski shows that

$$u(z) = \left(\frac{|C_z|}{v_B}\right)^{1/N}$$

is concave on $B - B$. It takes values in $[0,1]$ and $u(0) = 1$. The overlap volume is measurable (indeed continuous), and Fubini gives

$$\int_{B-B} u(z)^N\,dz = v_B.$$

Apply Lemma 2 with $d = p = N$ to $q = u^N/v_B$. Since $\mathrm{h}(Z) = \log v_B$, its conclusion is

$$\mathrm{h}(q) - \log v_B \le N \sum_{j=1}^{N} \frac{1}{N+j} = F(N).$$

□

3 Differences of scaled simplices

Let

$$\Delta_m = \left\{ x \in \mathbb{R}^m : x_i \ge 0,\ \sum_{i=1}^{m} x_i \le 1 \right\}, \qquad |\Delta_m| = \frac{1}{m!}.$$

Throughout this section U, V are independent uniform random vectors on Δ_m. In particular $\mathrm{h}(U) = -\log(m!)$.

Lemma 4. *Suppose $a \ge b > 0$, and let $g_{a,b}$ be the density of $aU - bV$. Put $q = 1 - b/a$ and $w_j = \binom{m}{j} q^j (1-q)^{m-j}$ for $0 \le j \le m$. For every real $p > 0$,*

$$\int_{\mathbb{R}^m} g_{a,b}(z)^p \, dz = \left(\frac{m!}{a^m}\right)^{p-1} \sum_{j=0}^{m} w_j \prod_{\ell=1}^{m-j} \frac{m+\ell}{mp+\ell}. \tag{9}$$

An empty product equals 1. *Consequently, for $J \sim \mathrm{Bin}(m, q)$,*

$$\mathrm{h}(aU - bV) = -\log(m!) + m \log a + m(\mathbb{E}H_{2m-J} - H_m). \tag{10}$$

In particular,

$$\mathrm{h}(U - V) - \mathrm{h}(U) = F(m). \tag{11}$$

Proof. For $z \in \mathbb{R}^m$, set

$$P(z) = \sum_{i=1}^{m} \max(z_i, 0), \qquad M(z) = \sum_{i=1}^{m} \max(-z_i, 0),$$

and $R(z) = \max\{0, \min(a - P(z), b - M(z))\}$. The conditions $x \in a\Delta_m$ and $x - z \in b\Delta_m$ are equivalent to

$$x_i \ge \max(0, z_i), \qquad \sum_i x_i \le \min\left(a, b + \sum_i z_i\right).$$

Subtracting the lower corner leaves a simplex of size $R(z)$. The convolution formula therefore gives

$$g_{a,b}(z) = \frac{m!}{a^m b^m} R(z)^m. \tag{12}$$

For $0 < t < b$, in an orthant with k positive coordinates the condition $R(z) > t$ is $P(z) < a - t$ and $M(z) < b - t$. Its volume is $(a-t)^k (b-t)^{m-k}/(k!(m-k)!)$. Coordinate hyperplanes have zero volume, so summing over the $\binom{m}{k}$ orthants gives

$$|\{R > t\}| = \frac{1}{m!} \sum_{k=0}^{m} \binom{m}{k}^2 (a-t)^k (b-t)^{m-k}. \tag{13}$$

The level set is empty for $t \geq b$.

Expand $a - t = (a - b) + (b - t)$. The coefficient identity

$$\sum_{k=j}^{m} \binom{m}{k}^2 \binom{k}{j} = \binom{m}{j} \binom{2m-j}{m} \tag{14}$$

follows by writing $\binom{m}{k}\binom{k}{j} = \binom{m}{j}\binom{m-j}{k-j}$ and applying Vandermonde's identity to the remaining sum. Thus

$$|\{R > t\}| = \frac{1}{m!} \sum_{j=0}^{m} \binom{m}{j} \binom{2m-j}{m} (a-b)^j (b-t)^{m-j}.$$

Using layer integration and (7), with $L = m - j$, gives

$$\int R^{mp} = \frac{1}{m!} \sum_{j=0}^{m} \binom{m}{j} \binom{2m-j}{m} (a-b)^j b^{mp+m-j} \frac{(m-j)!}{\prod_{\ell=1}^{m-j}(mp+\ell)}.$$

Here the formula includes $L = 0$, since $mp \int_0^b t^{mp-1}\, dt = b^{mp}$. Multiply by $(m!/(a^m b^m))^p$ and use

$$\binom{2m-j}{m}(m-j)! = \prod_{\ell=1}^{m-j}(m+\ell), \qquad \frac{(a-b)^j b^{m-j}}{a^m} = q^j(1-q)^{m-j}.$$

This proves (9).

The density $g_{a,b}$ is bounded with compact support. On, for example, $1/2 \leq p \leq 3/2$, the functions $g_{a,b}^p |\log g_{a,b}|$ have a common integrable bound. Hence entropy is minus the derivative of (9) at $p = 1$. Each finite product equals 1 at $p = 1$, and its derivative there is

$$-m \sum_{\ell=1}^{m-j} \frac{1}{m+\ell} = -m(H_{2m-j} - H_m).$$

Since $\sum_j w_j = 1$, (10) follows. Taking $a = b = 1$ gives (11). □

Proposition 5. *For every $a, b > 0$,*

$$\mathrm{h}(aU - bV) \geq \mathrm{h}(U - V) + \frac{m}{2}(\log a + \log b). \tag{15}$$

The inequality is strict if $a \neq b$.

Proof. Exchanging a, b reflects the density, without changing entropy. Thus assume $a \geq b$ and put $q = 1 - b/a \in [0, 1)$. By Lemma 4, the difference between the two sides of (15) equals

$$\mathcal{R}_m(q) = \frac{m}{2}[-\log(1-q)] - m\mathbb{E}[H_{2m} - H_{2m-J}], \qquad J \sim \mathrm{Bin}(m, q). \tag{16}$$

For $0 \leq j \leq m$,

$$H_{2m} - H_{2m-j} \leq \frac{j}{2m} + \frac{j(j-1)}{4m^2}. \tag{17}$$

Indeed, sum over $k = 0, \ldots, j-1$ the inequalities

$$\frac{1}{2m-k} = \frac{1}{2m} + \frac{k}{2m(2m-k)} \leq \frac{1}{2m} + \frac{k}{2m^2}.$$

Using $\mathbb{E}J = mq$ and $\mathbb{E}[J(J-1)] = m(m-1)q^2$, we get

$$\begin{aligned}\mathcal{R}_m(q) &\geq \frac{m}{2}[-\log(1-q) - q] - \frac{m-1}{4}q^2 \\ &\geq \frac{q^2}{4} \geq 0.\end{aligned}$$

The second inequality follows by integrating $1/(1-t) \geq 1+t$ over $0 \leq t \leq q$, which gives $-\log(1-q) \geq q + q^2/2$. If $a \neq b$, then $q > 0$. □

4 Lifting a concave power

Proposition 6. *Let $K \subset \mathbb{R}^n$ be a convex body, let $m \geq 1$ be an integer, and let $\rho : K \to (0,\infty)$ be upper semicontinuous and concave. Assume $0 < c \leq \rho \leq C < \infty$. If*

$$A = \int_K \rho(x)^m \, dx, \qquad f(x) = \frac{\rho(x)^m \mathbf{1}_K(x)}{A},$$

then

$$\mathrm{h}(f * \widetilde{f}) - \mathrm{h}(f) \leq F(n+m) - F(m). \tag{18}$$

Proof. Consider the convex body

$$B_\rho = \left\{ (x,y) \in K \times \mathbb{R}^m : y_i \geq 0,\ \sum_i y_i \leq \rho(x) \right\}.$$

It is convex by concavity of ρ, closed by upper semicontinuity, bounded, and has nonempty interior because K does and $\rho \geq c$. Its volume is $A/m!$. If $Z = (X,Y)$ is uniform on B_ρ, then X has density f, and given $X = x$ the density of Y is

$$k_x(y) = \frac{m!}{\rho(x)^m} \mathbf{1}_{\rho(x)\Delta_m}(y).$$

In particular, putting $L = \int_K f(x) \log \rho(x) \, dx$,

$$\mathrm{h}(Z) = \mathrm{h}(f) + \mathrm{h}(U) + mL. \tag{19}$$

This also follows directly from $\mathrm{h}(Z) = \log(A/m!)$ and $\mathrm{h}(f) = \log A - mL$.

Take an independent copy $Z' = (X', Y')$, and write $S = X - X'$ and $T = Y - Y'$. Let $g = f * \widetilde{f}$ be the density of S. For $g(s) > 0$, the conditional density of T given $S = s$ is

$$\kappa_s(t) = \int_{\mathbb{R}^n} \frac{f(x) f(x-s)}{g(s)} \, g_{\rho(x),\rho(x-s)}(t) \, dx, \tag{20}$$

where the integral is restricted to $x, x-s \in K$. The joint density of (S,T) is $g(s)\kappa_s(t)$. For $g(s) = 0$, the choice of κ_s is immaterial.

All simplex-difference densities in (20) have a common compact support and are bounded by $m!/c^m$: this follows either by convolution or by (12). The same is true of κ_s. Since $-u \log u$ is concave, Jensen's inequality, followed by integration in t, gives

$$\mathrm{h}(\kappa_s) \geq \int \frac{f(x) f(x-s)}{g(s)} \, \mathrm{h}\big(g_{\rho(x),\rho(x-s)}\big) \, dx.$$

All integrals are absolutely integrable: a uniformly bounded density on a fixed compact set has a uniformly bounded integral of $|u \log u|$. Multiplying by $g(s)$ and using the determinant-one change of variables $(x, s) \mapsto (x, x - s)$, Proposition 5 yields

$$\begin{aligned} \int g(s)\mathrm{h}(\kappa_s)\, ds &\geq \iint f(x)f(x')\mathrm{h}\big(g_{\rho(x),\rho(x')}\big)\, dx\, dx' \\ &\geq \mathrm{h}(U - V) + mL. \end{aligned} \tag{21}$$

The entropy identity for the joint density $g(s)\kappa_s(t)$ is

$$\mathrm{h}(Z - Z') = \mathrm{h}(g) + \int g(s)\mathrm{h}(\kappa_s)\, ds. \tag{22}$$

It follows by expanding $\log(g\kappa_s)$ on its positive set and using Fubini. The preceding bounds justify this identity as an equality of finite real numbers. Subtract (19) from (22) and use (21) and (11):

$$\mathrm{h}(Z - Z') - \mathrm{h}(Z) \geq \mathrm{h}(g) - \mathrm{h}(f) + F(m).$$

Proposition 3, applied in dimension $n+m$, bounds the left side by $F(n+m)$. This proves (18). □

5 Approximation and entropy convergence

Lemma 7. *For each fixed integer $n \geq 1$,*

$$\lim_{m\to\infty} \big(F(m+n) - F(m)\big) = n \log 2.$$

Proof. A direct calculation gives

$$F(k+1) - F(k) = H_{2k} - H_k + \frac{1}{2(2k+1)}.$$

The harmonic-number difference is the Riemann sum $k^{-1}\sum_{j=1}^{k}(1 + j/k)^{-1}$, which converges to $\int_0^1 (1+t)^{-1}\, dt = \log 2$. Sum the displayed identity for $k = m, \ldots, m+n-1$. □

Lemma 8. *Let $K \subset \mathbb{R}^n$ be a convex body and let $V : K \to [0, R]$ be lower semicontinuous and convex. The probability density $f = e^{-V}\mathbf{1}_K / \int_K e^{-V}$ satisfies (1).*

Proof. For integers $m > R$, put

$$\rho_m(x) = 1 - \frac{V(x)}{m}, \qquad f_m(x) = \frac{\rho_m(x)^m \mathbf{1}_K(x)}{\int_K \rho_m^m}.$$

The function ρ_m is upper semicontinuous and concave, and $1 - R/m \leq \rho_m \leq 1$. Thus Proposition 6 applies. The convergence $(1 - t/m)^m \to e^{-t}$ is uniform on $[0, R]$. It follows that $f_m \to f$ uniformly, with common compact support and a common bound for all sufficiently large m. Writing $g_m = f_m * \widetilde{f}_m$ and $g = f * \widetilde{f}$, we have

$$\|g_m - g\|_\infty \leq \|f_m - f\|_\infty \|f_m\|_1 + \|f\|_1 \|f_m - f\|_\infty = 2\|f_m - f\|_\infty.$$

Their supports are contained in $K - K$. Since $t \mapsto -t \log t$ is continuous on every compact interval in $[0, \infty)$, both entropies converge. Passing to the limit in (18) and using Lemma 7 proves the claim. □

Lemma 9. *If V is proper, lower semicontinuous and convex and $0 < \int_{\mathbb{R}^n} e^{-V} < \infty$, then V attains a finite minimum. After subtracting that minimum, its sufficiently large sublevel sets $K_R = \{V \le R\}$ are convex bodies and*

$$\frac{e^{-V(x)}}{\int e^{-V}} \le Ae^{-c|x|} \quad (x \in \mathbb{R}^n) \tag{23}$$

for some $A, c > 0$.

Proof. Some sublevel set K_{R_0} has positive volume, hence has nonempty interior. Moreover, $|K_R| \le e^R \int e^{-V} < \infty$ for each finite R. A convex set with nonempty interior and finite volume is bounded. Indeed, if it contains a ball and is unbounded, its convexity produces infinitely many disjoint balls of half that radius by taking midpoints of the fixed ball with a suitably separated sequence of distant points. This contradicts finite volume. Thus K_{R_0} is compact. The lower semicontinuous proper function V attains a finite minimum there; outside K_{R_0} it is larger than R_0, so this is a global minimum.

Subtract the minimum, and choose a minimizer x_0 with $V(x_0) = 0$. Choose $R_0 > 0$ so that K_{R_0} has nonempty interior and choose $L > 0$ with $K_{R_0} \subset \{|x - x_0| < L\}$. For $|x - x_0| > L$ and $V(x) < \infty$, put $t = L/|x - x_0|$ and $y = (1 - t)x_0 + tx$. Then $y \notin K_{R_0}$ and

$$R_0 < V(y) \le tV(x).$$

Together with $V \ge 0$, this gives globally

$$V(x) \ge \frac{R_0}{L}|x - x_0| - R_0.$$

This implies (23) after adjusting A for the translation by x_0. Every K_R with $R \ge R_0$ is closed, bounded and contains K_{R_0}, hence is a convex body. □

Lemma 10. *Suppose probability densities v_j, v on $\mathbb{R}^d$ satisfy $v_j \to v$ almost everywhere and, for fixed $A, c > 0$, $v_j(x), v(x) \le Ae^{-c|x|}$. Then $\int |v_j \log v_j|$ and $\int |v \log v|$ are finite, and $\mathrm{h}(v_j) \to \mathrm{h}(v)$.*

Proof. On any fixed ball, $|t \log t|$ is bounded for $0 \le t \le A$. Outside a sufficiently large ball $Ae^{-c|x|} \le e^{-1}$. Since $-t \log t$ is increasing on $[0, e^{-1}]$, there

$$|v_j(x) \log v_j(x)| \le Ae^{-c|x|}(c|x| + |\log A|),$$

and the same bound holds for v. This is integrable. Apply dominated convergence, using continuity of $t \log t$ at zero. □

6 Conclusion

Proof of Theorem 1. Represent $f = e^{-V}/Z$ and, by Lemma 9, arrange $V \ge 0$. For sufficiently large integers R, put

$$K_R = \{V \le R\}, \qquad p_R = \int_{K_R} f, \qquad f_R = \frac{f\mathbf{1}_{K_R}}{p_R}.$$

We have $p_R \uparrow 1$. Lemma 8 gives

$$\mathrm{h}(f_R * \widetilde{f}_R) - \mathrm{h}(f_R) \le n \log 2. \tag{24}$$

It remains to remove the truncation with both entropies controlled. For all sufficiently large R, $p_R \ge 1/2$, so $f_R \le 2f$ and $f_R \to f$ pointwise. Lemma 9 supplies a common exponential envelope. Lemma 10 therefore gives $\mathrm{h}(f_R) \to \mathrm{h}(f)$ and finiteness of the latter.

Let $g = f * \widetilde{f}$ and $g_R = f_R * \widetilde{f}_R$. Then $g_R \le 4g$, and $g_R(z) \to g(z)$ for every z: the unnormalized convolution integrands increase to $f(x)f(x-z)$ and $p_R^2 \to 1$. If $f(x) \le Ae^{-c|x|}$, the triangle inequality yields

$$\begin{aligned} g(z) &\le A^2 \int e^{-c(|x|+|x-z|)}\,dx \\ &\le A^2 e^{-c|z|/2} \int e^{-c|x|/2}\,dx. \end{aligned}$$

Hence g_R, g also have a common exponential envelope. Lemma 10 gives $\mathrm{h}(g_R) \to \mathrm{h}(g)$ and finiteness of $\mathrm{h}(g)$. Taking $R \to \infty$ in (24) proves the bound.

For sharpness, let $E_1, \dots, E_n$ be independent variables with density $e^{-t}\mathbf{1}_{[0,\infty)}(t)$ and put $E = (E_1, \dots, E_n)$. Direct integration gives $\mathrm{h}(E_i) = 1$. The difference of two independent copies has density $e^{-|t|}/2$ and entropy $1 + \log 2$. Product densities therefore give

$$\mathrm{h}(E - E') - \mathrm{h}(E) = n \log 2.$$

If A is invertible, change of variables gives $\mathrm{h}(AX + b) = \mathrm{h}(X) + \log|\det A|$. The two determinant terms in the entropy increment cancel, establishing the asserted affine equality examples. □